ECE 121B Midterm formula sheet

Carriers, statistics and transport — Professor Emaminejad / Lectures 1–8

Reference values at 300 K

L1 p25; L3 p8–10

$q = 1.602 \times 10^{-19}$ C, $k_B = 8.617 \times 10^{-5}$ eV/K

$V_T = \dfrac{k_B T}{q} = 25.85$ mV, $E_{g,\mathrm{Si}} = 1.12$ eV

$\varepsilon_0 = 8.85 \times 10^{-14}$ F/cm, $\varepsilon_{\mathrm{Si}} = 11.7\varepsilon_0$

$N_C = 2.82 \times 10^{19}$, $N_V = 1.83 \times 10^{19}$ cm^{-3}

$n_i \approx 10^{10}$ cm^{-3} (rounded course value)

Use $k_B = 1.381 \times 10^{-23}$ J/K with q in coulombs in V_T. Match energy units in every exponential.

1. Occupation and carrier statistics

L2 p15,30–32; L3 p6–12

$$f(E) = \frac{1}{1 + \exp[(E - E_F)/(k_B T)]}$$

$f \approx e^{-(E-E_F)/(k_B T)}$ $(E - E_F \geq 3k_B T)$

$1 - f \approx e^{-(E_F-E)/(k_B T)}$ $(E_F - E \geq 3k_B T)$

$g_C(E) \propto \sqrt{E - E_C}$, $g_V(E) \propto \sqrt{E_V - E}$

$$n = \int_{E_C}^{\infty} g_C f \, dE, \quad p = \int_{-\infty}^{E_V} g_V (1 - f) \, dE$$

$$N_{C,V} = 2 \left(\frac{2\pi m^*_{n,p,\mathrm{DOS}} k_B T}{h^2} \right)^{3/2} \propto T^{3/2}$$

$n = N_C e^{-(E_C - E_F)/(k_B T)}$, $p = N_V e^{-(E_F - E_V)/(k_B T)}$

$n_i = \sqrt{N_C N_V}\, e^{-E_g/(2k_B T)}$, $np = n_i^2$

$$E_i = \frac{E_C + E_V}{2} + \frac{k_B T}{2} \ln \frac{N_V}{N_C}$$

2. Doping and charge neutrality

L2 p23–26; L3 p13–20

$n + N_A = p + N_D$, $\Delta N = N_D - N_A$

$$n = \frac{\Delta N + \sqrt{\Delta N^2 + 4n_i^2}}{2}, \quad p = \frac{n_i^2}{n}$$

n-type: $n \approx N_D - N_A$, $p \approx \dfrac{n_i^2}{n}$

p-type: $p \approx N_A - N_D$, $n \approx \dfrac{n_i^2}{p}$

$E_V + 3k_B T \leq E_F \leq E_C - 3k_B T$ (Boltzmann)

Full ionization assumed. Majority-carrier approximations need net doping $\gg n_i$. Freeze-out violates full ionization.

At 300 K, degeneracy starts near $N_D > 1.6 \times 10^{18}$ or $N_A > 9.1 \times 10^{17}$ cm^{-3}. Heavy n$^+$: $E_F \approx E_C$; heavy p$^+$: $E_F \approx E_V$.

$$\Delta E_g \approx 3.5 \times 10^{-8} N^{1/3} \frac{300}{T} \text{ eV}$$

Bandgap narrowing estimate: N in cm^{-3}, $N \geq 10^{18}$ cm^{-3}.

3. Drift, diffusion and conductivity

L3 p26–41; L4 p18–22

$v_n = -\mu_n \mathcal{E}$, $v_p = +\mu_p \mathcal{E}$

$$J_n = qn\mu_n \mathcal{E} + qD_n \frac{dn}{dx}$$

$$J_p = qp\mu_p \mathcal{E} - qD_p \frac{dp}{dx}, \quad J = J_n + J_p$$

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T \quad \text{(Einstein relation)}$$

$$\sigma = q(n\mu_n + p\mu_p), \quad \rho_{\mathrm{res}} = \frac{1}{\sigma}, \quad R = \frac{\rho_{\mathrm{res}} L}{A}$$

$$\mathcal{E} = -\frac{d\psi}{dx} = \frac{1}{q}\frac{dE_C}{dx}$$

Equilibrium: $\mathcal{E} = -\dfrac{V_T}{n}\dfrac{dn}{dx}$

$$\psi(x_2) - \psi(x_1) = V_T \ln \frac{n(x_2)}{n(x_1)}$$

Electrons drift opposite $\mathcal{E}$; holes drift with $\mathcal{E}$. Conventional drift current is along $\mathcal{E}$ for both. In $(1/q)dE_C/dx$, use E_C in joules.

4. Continuity and recombination

L4 p29–30; L5 p5–17

Pair generation: $\Delta n = \Delta p$

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G - R$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G - R$$

$$R_{\mathrm{net}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}$$

$n_1 = n_i e^{(E_T - E_i)/(k_B T)}$, $p_1 = n_i e^{(E_i - E_T)/(k_B T)}$

$R_{\mathrm{net}} > 0$: net recombination. Low injection: $|\Delta n| \ll n_0$ in n-type; $|\Delta p| \ll p_0$ in p-type.

$$\frac{\partial \Delta p}{\partial t} = D_p \frac{\partial^2 \Delta p}{\partial x^2} - \frac{\Delta p}{\tau_p} + G_L$$

$L_p = \sqrt{D_p \tau_p}$, $L_n = \sqrt{D_n \tau_n}$

No current/source: $\Delta p(t) = \Delta p(0) e^{-t/\tau_p}$

Steady, $G_L = 0$: $\Delta p(x) = \Delta p(0) e^{-x/L_p}$

Minority-diffusion forms assume a field-free neutral region and low injection. Swap $n \leftrightarrow p$ for electrons in p-type material.

5. Quasi-Fermi levels

L5 p21–23

$n = n_i e^{(F_n - E_i)/(k_B T)}$, $p = n_i e^{(E_i - F_p)/(k_B T)}$

$F_n = E_i + k_B T \ln \dfrac{n}{n_i}$, $F_p = E_i - k_B T \ln \dfrac{p}{n_i}$

$$np = n_i^2 \exp\left(\frac{F_n - F_p}{k_B T} \right)$$

Boltzmann relations. At equilibrium $F_n = F_p = E_F$ and $np = n_i^2$.

Scattering and mobility

L4 p4–8 (repeats L3 p33–37)

$\mu_{\mathrm{phonon}} \propto T^{-3/2}$, $\mu_{\mathrm{impurity}} \propto \dfrac{T^{3/2}}{N_D + N_A}$

$$\mu = \frac{q\tau}{m^*}, \qquad \frac{1}{\mu} = \frac{1}{\mu_{\mathrm{phonon}}} + \frac{1}{\mu_{\mathrm{impurity}}}$$

Matthiessen's rule: the mechanism with the lower mobility dominates. More heat increases phonon scattering; faster carriers are deflected less by fixed ions.

ECE 121B Midterm formula sheet

Junctions, contacts and diode current — Professor Emaminejad / Lectures 1–8

6. Electrostatics and abrupt PN junctions

L5 p24–25; L6 p17–29

$$\rho_{\text{ch}} = q(p - n + N_D^+ - N_A^-)$$

$$\frac{d\mathcal{E}}{dx} = \frac{\rho_{\text{ch}}}{\varepsilon_s}, \quad \mathcal{E} = -\frac{d\psi}{dx}, \quad \frac{d^2\psi}{dx^2} = -\frac{\rho_{\text{ch}}}{\varepsilon_s}$$

$$V_{bi} = V_T \ln\frac{N_A N_D}{n_i^2}, \quad U = V_{bi} - V_A > 0$$

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)U}$$

$$x_p = \frac{W N_D}{N_A + N_D}, \quad x_n = \frac{W N_A}{N_A + N_D}$$

$$N_A x_p = N_D x_n, \quad W = x_p + x_n$$

$$|\mathcal{E}_{\max}| = \frac{qN_A x_p}{\varepsilon_s} = \frac{qN_D x_n}{\varepsilon_s} = \frac{2U}{W}$$

$$\text{p-side: } \mathcal{E}(x) = -\frac{qN_A}{\varepsilon_s}(x + x_p)$$

$$\text{n-side: } \mathcal{E}(x) = +\frac{qN_D}{\varepsilon_s}(x - x_n)$$

$$|Q| = qAN_A x_p = qAN_D x_n$$

$$\text{One-sided: } W \approx \sqrt{\frac{2\varepsilon_s U}{qN_{\text{light}}}}$$

$$\text{p}^+\text{n / n}^+\text{p: } qV_{bi} \approx \frac{E_g}{2} + k_B T \ln\frac{N_{\text{light}}}{n_i}$$

$V_A > 0$ is PN forward bias. Depletion has fixed dopant charge and negligible mobile charge. Ordinary V_{bi} assumes nondegenerate statistics; the heavy-side approximation pins E_F at its band edge.

7. Metal-semiconductor contacts

L5 p27–39; L6 p4–8

$$\phi_{Bn} = \phi_M - \chi, \quad \phi_{Bp} = \frac{E_g}{q} - \phi_{Bn} \quad \text{(ideal)}$$

$$\text{n-type: } V_{bi} = \phi_{Bn} - V_T \ln\frac{N_C}{N_D}$$

$$\text{p-type: } V_{bi} = \phi_{Bp} - V_T \ln\frac{N_V}{N_A}$$

$$W_n = \sqrt{\frac{2\varepsilon_s(V_{bi} - V_A)}{qN_D}}$$

$$W_p = \sqrt{\frac{2\varepsilon_s(V_{bi} + V_A)}{qN_A}}$$

ϕ and χ are energy/q in volts; eV energies have the same numerical value. P-type polarity (L6 p8): $V_A > 0$ is reverse bias.

NiSi (L5 p33): electron barrier 0.65 eV; hole barrier 0.47 eV; work function 4.65 eV. Real interfaces can depart from ideal Schottky-Mott values.

Symbols and units

Course notation

n, p: electron/hole densities; N_D, N_A: donor/acceptor densities. A: area; μ: mobility; D: diffusivity; τ: lifetime; L: diffusion length.

E_F: equilibrium Fermi level; E_i: intrinsic level; E_C, E_V: band edges; $\mathcal{E}$: electric field; ϕ_B: barrier energy/q.

ρ_{ch}: charge density (C/cm^3); ρ_{res}: resistivity (Ω·cm). J: A/cm^2; $I = AJ$: amperes.

8. Depletion capacitance and C–V

L6 p9–11

$$C = \frac{\varepsilon_s A}{W}, \quad i_{\text{ac}} = C\frac{dv_{\text{ac}}}{dt}$$

$$\frac{1}{C^2} = \frac{2(V_{bi} - V_A)}{q\varepsilon_s A^2 N}$$

$$m = \frac{d(1/C^2)}{dV_A} = -\frac{2}{q\varepsilon_s A^2 N}$$

$$N = -\frac{2}{q\varepsilon_s A^2 m}, \quad \text{voltage intercept} = V_{bi}$$

Slope sign uses n-type Schottky or one-sided PN forward-bias convention. A is area. Use consistent cm/F units.

9. Ideal PN diode

L7 p6–20

$$n_{p0} = \frac{n_i^2}{N_A}, \quad p_{n0} = \frac{n_i^2}{N_D}$$

$$n_p(-x_p) = n_{p0}e^{V_A/V_T}, \quad p_n(x_n) = p_{n0}e^{V_A/V_T}$$

$$J = J_0\left(e^{V_A/V_T} - 1\right), \quad I = AJ$$

$$J_0 = qn_i^2\left(\frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D}\right)$$

$$\Delta p_n(x_n + x) = p_{n0}\left(e^{V_A/V_T} - 1\right)e^{-x/L_p}$$

$$x \geq 0, \quad \Delta V_{A,\text{decade}} = V_T \ln 10 \approx 59.5 \text{ mV}$$

At 300 K. Assumes long neutral regions, steady state, low injection, full ionization, nondegenerate statistics and no depletion-region recombination. Ideal reverse current tends to $-J_0$ before breakdown.

10. Nonideal current and breakdown

L8 p4–16

$$\text{Reverse R–G: } I_{\text{R-G}} \approx -\frac{qAn_i W}{2\tau_0}$$

$$\tau_0 = \frac{\tau_p n_1 + \tau_n p_1}{2n_i}$$

$$\text{Forward R–G: } I_{\text{R-G}} \propto n_i W e^{V_A/(2V_T)}$$

$$\text{Series resistance: } V_A = V_J + IR_S$$

$$V_{\text{BR}} \approx \frac{\varepsilon_s \mathcal{E}_{\text{CR}}^2}{2qN_{\text{light}}} - V_{bi}$$

Critical-field estimate for a one-sided junction. If $V_{\text{BR}} \gg V_{bi}$, drop V_{bi}. This is not a breakdown-current law.

Avalanche: impact ionization; $\mathcal{E}_{\text{CR}} \sim 10^5$–$10^6$ V/cm; V_{BR} generally rises with temperature.

Zener: heavy doping, $W \lesssim 10$ nm, $\mathcal{E}_{\text{CR}} \sim 10^6$ V/cm; typically $V_{\text{BR}} < 5$ V and falls with temperature.

High injection and series resistance invalidate simple ideal-current assumptions.

Energy and sign checks

Before substituting

Use joules with k_B in J/K when explicitly dividing energy by q. For eV energies, $k_B = 8.617 \times 10^{-5}$ eV/K; energy/q has the same numerical value in volts.

Electron velocity and electron current have opposite signs. Label electric field, voltage reference and positive current direction before using a transport formula.